

DATA 260 - Homework 3

Section 0. Personal Configuration and Domain

Value	Configuration
SID4	9486
PORT_BASE	8486
PREFIX	s9486
SEED	9486
VERIFY_SEED	269486
DOMAIN_ID	6
Assigned Domain	Rental housing listings
Hardware	Apple MacBook Air with Apple M1 chip and 8 GB unified memory
Local model	Not applicable - Part 2 uses the sentence-transformers/all-MiniLM-L6-v2 embedding model (Hugging Face); no local LLM was required for this homework
Tagged commit hash (<code>git rev-parse HEAD</code> after tagging <code>hw3</code>)	TODO_RUN_MANUALLY
GitHub repository	https://github.com/Kavan1804/data260-9486 (confirm Sbnikitha and supriyaselvanganesan have access)

The six configuration values above are computed in `config.py` and were carried over unchanged from `Homework1/README.md` , since the assignment specifies they “stay fixed for the rest of the

semester.”

Reproducible run instructions

```
cd Homework3
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

# Part 1: auth app
python3 main.py
# open http://localhost:8486/

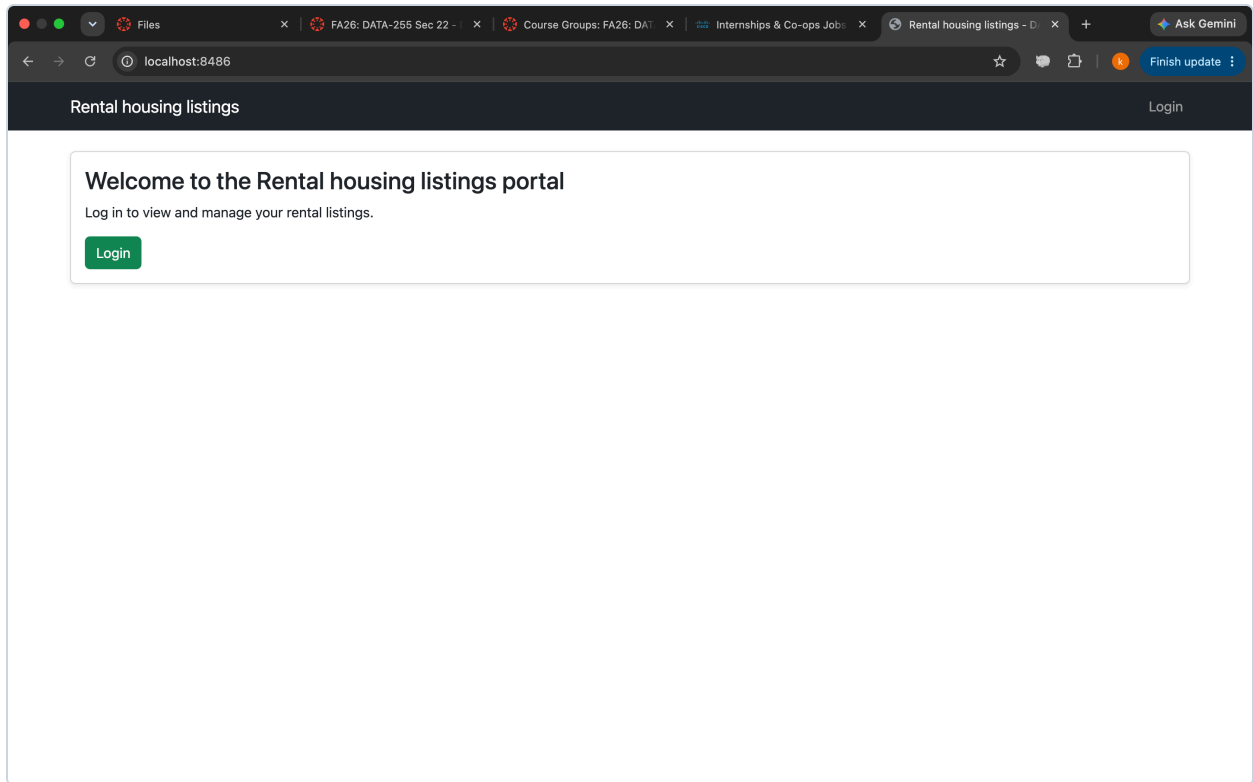
# Part 2: retrieval experiments
python rag/corpus_prep.py
python rag/build_manifest.py
python -m rag.run_experiments
python -m rag.compute_metrics

# Self-check
COMMIT_HASH=$(git rev-parse HEAD) python3 scripts/verify_hw03.py
```

Part 1. Authentication App

Routes

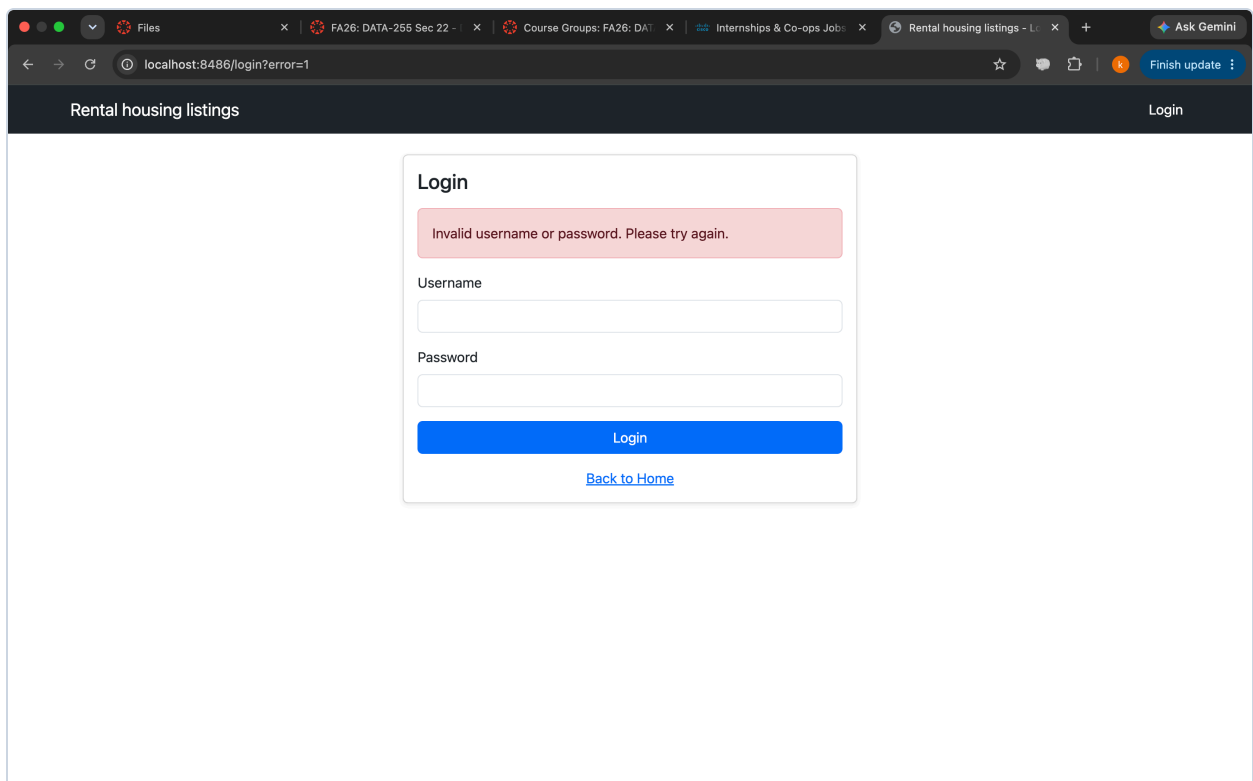
- GET / - domain welcome message, login link or dashboard/logout links depending on session state.
- GET /login / POST /login - login form; invalid credentials show a Bootstrap alert.
- GET /dashboard - protected; requires an active, non-idle session.
- GET /logout - clears the session and redirects home.



GET / while logged out

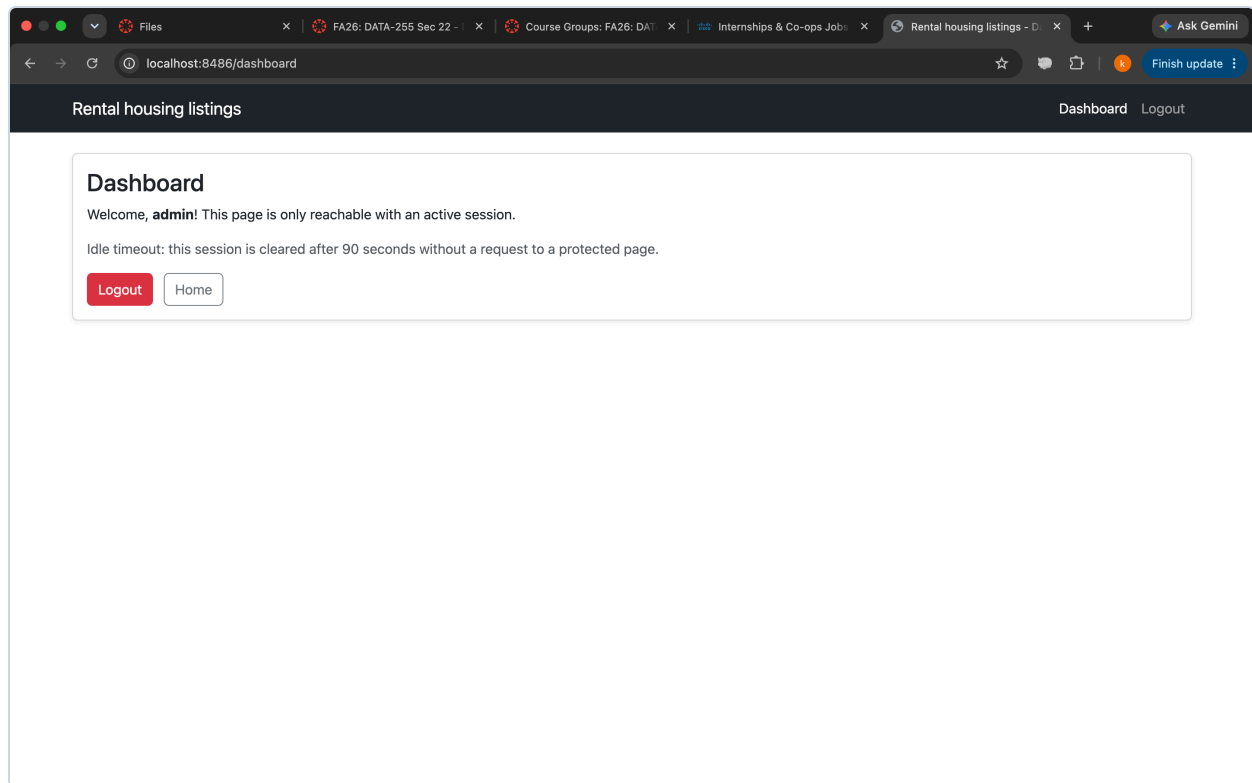
GET / while logged out - Login link shown in navbar, welcome card with a Login button.

[SCREENSHOT PLACEHOLDER: GET / while logged in - still needed. Log in, then screenshot the home page, which should show “You are logged in as admin” plus Dashboard/Logout buttons.]



GET /login showing the invalid-credentials Bootstrap alert

Submitting the wrong password redirects to `/login?error=1`, which renders the red Bootstrap alert “Invalid username or password. Please try again.”



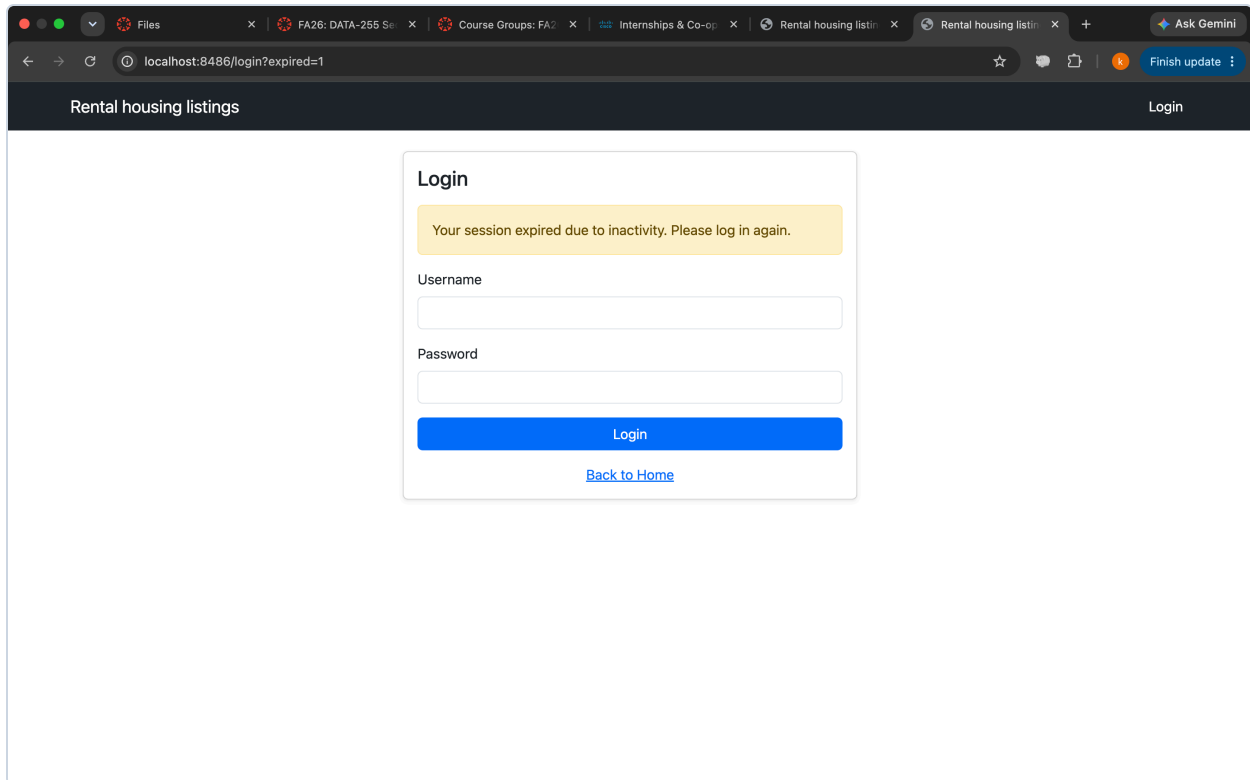
GET /dashboard after a successful login

Dashboard after logging in as `admin`: welcome message, idle-timeout note, Logout/Home buttons.

[SCREENSHOT PLACEHOLDER: GET /logout redirecting back to / logged out - still needed. Click Logout from the dashboard and screenshot the resulting home page (should look like the logged-out home screenshot above, reached via the logout redirect).]

[SCREENSHOT PLACEHOLDER: templates/ directory listing (index.html, login.html, dashboard.html) - still needed. A Finder/VS Code sidebar screenshot of Homework3/templates/ is enough.]

Session cookie evidence (Secure, HttpOnly, SameSite)



Session expired after idle timeout

After logging in and then waiting more than 90 seconds without visiting a protected route, requesting `/dashboard` again redirects to `/login?expired=1`, which shows the yellow “Your session expired due to inactivity” alert - proving the idle session could not reach `/dashboard`.

[SCREENSHOT PLACEHOLDER: `/dashboard` redirecting to `/login` after logout (not idle-timeout) - still needed. Log in, click Logout, then request `/dashboard` directly to show the cleared session is also rejected, distinct from the idle-timeout case above.]

auth.py

The full router implementation, as required by the assignment (“Include only the auth.py code at the bottom”):

```
import time
from typing import Optional

from fastapi import APIRouter, Form, Request
from fastapi.responses import RedirectResponse
from fastapi.templating import Jinja2Templates
from starlette.status import HTTP_302_FOUND

from config import DOMAIN_NAME, IDLE_TIMEOUT_SECONDS

# Behaves like a mini FastAPI app; wired into main.py with app.include_router()
```

```

router = APIRouter()

templates = Jinja2Templates(directory="templates")

# Hardcoded credentials for local manual verification only.
# A real deployment would check a user database with hashed passwords.
VALID_USERNAME = "admin"
VALID_PASSWORD = "password"

def _session_status(request: Request):
    """Return ("active", user), ("expired", None), or ("none", None).

    Idle timeout is not something Starlette's SessionMiddleware tracks on its
    own (max_age is a fixed cookie lifetime, not an idle window), so the last
    active time is stored inside the session and checked on every request to
    a protected route.
    """

    user = request.session.get("user")
    if not user:
        return "none", None

    last_active = request.session.get("last_active")
    if last_active is None or time.time() - last_active > IDLE_TIMEOUT_SECONDS:
        request.session.clear()
        return "expired", None

    request.session["last_active"] = time.time() # sliding idle window
    return "active", user

@router.get("/")
def home(request: Request):
    _, user = _session_status(request)
    return templates.TemplateResponse(
        request,
        "index.html",
        {"user": user, "domain_name": DOMAIN_NAME},
    )

@router.get("/login")
def login_page(request: Request, error: Optional[str] = None, expired: Optional[str] =
    None):
    return templates.TemplateResponse(
        request,
        "login.html",
        {
            "domain_name": DOMAIN_NAME,
            "show_invalid_alert": error == "1",

```

```

        "show_expired_alert": expired == "1",
    },
)

@router.post("/login")
def login(request: Request, username: str = Form(...), password: str = Form(...)):
    if username == VALID_USERNAME and password == VALID_PASSWORD:
        request.session["user"] = username
        request.session["last_active"] = time.time()
        return RedirectResponse(url="/dashboard", status_code=HTTP_302_FOUND)

    # Invalid credentials: redirect back with a flag the template turns into
    # a Bootstrap alert, instead of silently reloading the empty form.
    return RedirectResponse(url="/login?error=1", status_code=HTTP_302_FOUND)

@router.get("/dashboard")
def dashboard(request: Request):
    status, user = _session_status(request)

    if status == "expired":
        return RedirectResponse(url="/login?expired=1", status_code=HTTP_302_FOUND)
    if status == "none":
        return RedirectResponse(url="/login", status_code=HTTP_302_FOUND)

    return templates.TemplateResponse(
        request,
        "dashboard.html",
        {
            "user": user,
            "domain_name": DOMAIN_NAME,
            "idle_timeout": IDLE_TIMEOUT_SECONDS,
        },
    )

@router.get("/logout")
def logout(request: Request):
    request.session.clear()
    return RedirectResponse(url="/", status_code=HTTP_302_FOUND)

```

Part 2. Retrieval-Only RAG - Three Chunking Techniques

Corpus: 5 public California/federal rental-housing law documents totaling 649,786 bytes (see `SOURCES.md` and `CORPUS_MANIFEST.json`). Note: `hud_fair_market_rents.txt` downloaded as 0

bytes (a broken source URL) - see the caveat in `METRICS.md` ; the numbers below reflect the corpus as actually downloaded, including that gap.

```

$ FIRST
• (.venv) (base) kavan@Kavans-MacBook-Air Homework3 % python rag/corpus_prep.py
Downloading https://leginfo.ca.gov/faces/codes_displaySection.xhtml?sectionNum=1950.5.&lawCode=CIV -> /Users/kavan/Desktop/Data 260 - DS/Homework1/data260-9486/Homework3/rag/corpus/ca_civil_code_1950_5_security_deposit.txt
192916 bytes
Downloading https://leginfo.ca.gov/faces/codes_displaySection.xhtml?sectionNum=1946.2.&lawCode=CIV -> /Users/kavan/Desktop/Data 260 - DS/Homework1/data260-9486/Homework3/rag/corpus/ca_tenant_protection_act_ab1482.txt
192922 bytes
Downloading https://www.justice.gov/crt/fair-housing-act-1 -> /Users/kavan/Desktop/Data 260 - DS/Homework1/data260-9486/Homework3/rag/corpus/fair_housing_act.txt
98818 bytes
Downloading https://leginfo.ca.gov/faces/codes_displaySection.xhtml?sectionNum=1946.&lawCode=CIV -> /Users/kavan/Desktop/Data 260 - DS/Homework1/data260-9486/Homework3/rag/corpus/ca_tenant_notice_requirements.txt
165130 bytes
Downloading https://www.huduser.gov/portal/datasets/fmr.html -> /Users/kavan/Desktop/Data 260 - DS/Homework1/data260-9486/Homework3/rag/corpus/hud_fair_market_rents.txt
0 bytes

Total corpus size: 649786 bytes (634.6 KB)
```

corpus_prep.py console output

`python rag/corpus_prep.py` downloading the four working sources plus the broken 0-byte HUD file.

```

• (.venv) (base) kavan@Kavans-MacBook-Air Homework3 % python rag/build_manifest.py

{
  "total_bytes": 649786,
  "files": [
    {
      "filename": "ca_civil_code_1950_5_security_deposit.txt",
      "byte_size": 192916,
      "sha256": "5521f50bf093923ab9842666196059b50a4d538e8753bc56667888c464595f81"
    },
    {
      "filename": "ca_tenant_notice_requirements.txt",
      "byte_size": 165130,
      "sha256": "1cd5c88a53d7fb20d0e9c1ee9c0f14971a39e3c1bb6d4d647c6806e4361abded"
    },
    {
      "filename": "ca_tenant_protection_act_ab1482.txt",
      "byte_size": 192922,
      "sha256": "109ed3ed6fd75436fe45517fe7417e27efc0d7da06eb590039c3f7f922e6bc0d"
    },
    {
      "filename": "fair_housing_act.txt",
      "byte_size": 98818,
      "sha256": "ad8d7724724adc5662ddaecae8971f9ba85bab95d67b5fd74697051f1d61d0a"
    },
    {
      "filename": "hud_fair_market_rents.txt",
      "byte_size": 0,
      "sha256": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
    }
  ]
}

Wrote /Users/kavan/Desktop/Data 260 - DS/Homework1/data260-9486/Homework3/reports/hw03/CORPUS_MANIFEST.json
```

build_manifest.py console output

`python rag/build_manifest.py` computing real byte sizes and SHA-256 hashes into `CORPUS_MANIFEST.json`.

Code snippets

Token-based chunking (`TokenTextSplitter`)

```

from rag.pipelines import build_nodes, build_index, retrieve_and_score, load_corpus,
    get_embed_model

embed_model = get_embed_model()
documents = load_corpus("rag/corpus")
nodes = build_nodes(documents, "token", embed_model=embed_model)
index = build_index(nodes, embed_model=embed_model)
rows, latency_ms = retrieve_and_score(index, "token", "<one of the five questions>",
    k=5, embed_model=embed_model)

```

Output per query: the query embedding dimension (384 for `all-MiniLM-L6-v2`) and its first 8 values, the query-vector and stacked doc-vectors shapes, then a `rank | store_score | cosine_sim | chunk_len | preview` line per retrieved node - written to `raw/token.jsonl`.

Semantic chunking (`SemanticSplitterNodeParser`)

```

nodes = build_nodes(documents, "semantic", embed_model=embed_model)
index = build_index(nodes, embed_model=embed_model)
rows, latency_ms = retrieve_and_score(index, "semantic", "<same question>", k=5,
    embed_model=embed_model)

```

Written to `raw/semantic.jsonl`.

Sentence-window chunking (`SentenceWindowNodeParser`)

```

nodes = build_nodes(documents, "sentence_window", embed_model=embed_model)
index = build_index(nodes, embed_model=embed_model)
rows, latency_ms = retrieve_and_score(index, "sentence_window", "<same question>", k=5,
    embed_model=embed_model)

```

Written to `raw/sentence_window.jsonl`.

```
DATA 260 Homework 3 Implementation
RICS.md > # HW3 Metrics - Chunking Technique Com... Preview Markdown

Problems Output Debug Console Terminal Ports Python - Homework3 + - [] ... ^ x

(.venv) (base) kavan@Kavans-MacBook-Air Homework3 % python -m rag.compute_metrics

| Technique | Chunks | Avg chunk length | Top-1 cosine | Meangk cosine | Recall@k | Mean retrieval latency (ms) |
|-----|-----|-----|-----|-----|-----|-----|
| Token | 2365 | 307.1 | 0.5949 | 0.5158 | 1.00 | 53.91 |
| Semantic | 18 | 36072.1 | 0.3344 | 0.1951 | 1.00 | 15.33 |
| Sentence window | 236 | 2751.3 | 0.5819 | 0.4867 | 0.80 | 14.03 |

Confidently scored misses found: 7
{
  "technique": "token",
  "query_id": "q1",
  "query": "What is the maximum security deposit a California landlord can charge for an unfurnished unit?",
  "cosine_sim": 0.5242845762867607,
  "preview": "the landlord shall provide the tenant with a written statement, on or before the date the lease is signed, of the amount of the higher security and an explanati",
  "expected_answer": "One month's rent"
}
{
  "technique": "token",
  "query_id": "q2",
  "query": "Under California's Tenant Protection Act (AB 1482), what is the general annual cap on rent increases for a covered unit?",
  "cosine_sim": 0.5803090368176448,
  "preview": "existing prior to July 1, 2022, by written notice to the tenant no later than August 1, 2022, or as an addendum to the lease or rental agreement.</p><p style='m",
  "expected_answer": "5% plus local CPI, capped at 10% total"
}
{
  "technique": "token",
  "query_id": "q3",
  "query": "Which federal law prohibits housing discrimination based on race, color, religion, sex, national origin, familial status, and disability?",
  "cosine_sim": 0.6439787308769069,
  "preview": "Credit Opportunity Act</a> have alleged discrimination based on race or color. Some of the Department's cases have also alleged that municipalities an other lo",
  "expected_answer": "The Fair Housing Act of 1968"
}
{
  "technique": "token",
  "query_id": "q5",
  "query": "In California, how many days' written notice must a landlord give to end a month-to-month tenancy after the tenant has lived there for one year or more?",
  "cosine_sim": 0.7835026172216681,
  "preview": "existing prior to July 1, 2022, by written notice to the tenant no later than August 1, 2022, or as an addendum to the lease or rental agreement.</p><p style='m",
  "expected_answer": "60 days"
}
{
  "technique": "semantic",
  "query_id": "q3",
  "query": "Which federal law prohibits housing discrimination based on race, color, religion, sex, national origin, familial status, and disability?",
  "cosine_sim": 0.6254711945786787,
  "preview": "Where force or threat of force is used to deny or interfere with fair housing rights, the Department of Justice may institute <a href='\"http://www.jus",
  "expected_answer": "The Fair Housing Act of 1968"
}
0
```

compute_metrics.py console output - summary table and confident misses (1/2)

```
corpus_prep.py U DATA 260 Homework 3 impl... METRICS.md U
DATA 260 Homework 3 Implementation
RICS.md > # HW3 Metrics - Chunking Technique Com... Preview Markdown

Problems Output Debug Console Terminal Ports Python - Homework3 + - [] ... ^ x

"expected_answer": "5% plus local CPI, capped at 10% total"
}
{
  "technique": "token",
  "query_id": "q3",
  "query": "Which federal law prohibits housing discrimination based on race, color, religion, sex, national origin, familial status, and disability?",
  "cosine_sim": 0.6439787308769069,
  "preview": "Credit Opportunity Act</a> have alleged discrimination based on race or color. Some of the Department's cases have also alleged that municipalities an other lo",
  "expected_answer": "The Fair Housing Act of 1968"
}
{
  "technique": "token",
  "query_id": "q5",
  "query": "In California, how many days' written notice must a landlord give to end a month-to-month tenancy after the tenant has lived there for one year or more?",
  "cosine_sim": 0.7835026172216681,
  "preview": "existing prior to July 1, 2022, by written notice to the tenant no later than August 1, 2022, or as an addendum to the lease or rental agreement.</p><p style='m",
  "expected_answer": "60 days"
}
{
  "technique": "semantic",
  "query_id": "q3",
  "query": "Which federal law prohibits housing discrimination based on race, color, religion, sex, national origin, familial status, and disability?",
  "cosine_sim": 0.6254711945786787,
  "preview": "Where force or threat of force is used to deny or interfere with fair housing rights, the Department of Justice may institute <a href='\"http://www.jus",
  "expected_answer": "The Fair Housing Act of 1968"
}
{
  "technique": "sentence_window",
  "query_id": "q3",
  "query": "Which federal law prohibits housing discrimination based on race, color, religion, sex, national origin, familial status, and disability?",
  "cosine_sim": 0.533117793545332,
  "preview": "Such discrimination can be based either upon the country of an individual's birth or where his or her ancestors originated. ",
  "expected_answer": "The Fair Housing Act of 1968"
}
{
  "technique": "sentence_window",
  "query_id": "q5",
  "query": "In California, how many days' written notice must a landlord give to end a month-to-month tenancy after the tenant has lived there for one year or more?",
  "cosine_sim": 0.7871258930518512,
  "preview": "California law also provides that after all of the tenants have continuously and lawfully occupied the property for 12 months or more or at least one of the ten",
  "expected_answer": "60 days"
}
}

Wrote /Users/Kavan/Desktop/Data 260 - DS/Homework1/data260-9486/Homework3/reports/hw03/raw/summary_metrics.json
Paste the table above into reports/hw03/METRICS.md.
(.venv) (base) kavan@Kavans-MacBook-Air Homework3 %
```

compute_metrics.py console output - confident misses (2/2)

`python -m rag.compute_metrics` reading `raw/token.jsonl`, `raw/semantic.jsonl`, and `raw/sentence_window.jsonl`, printing the summary table above and all 7 confidently scored misses, then writing `raw/summary_metrics.json`.

Retrieval quality comparison

Technique	Chunks	Avg chunk length	Top-1 cosine	Mean@k cosine	Recall@k	Mean retrieval latency (ms)
Token	2365	307.1	0.5949	0.5158	1.00	53.91
Semantic	18	36072.1	0.3344	0.1951	1.00	15.33
Sentence window	236	2751.3	0.5819	0.4867	0.80	14.03

(Source: `raw/summary_metrics.json`, computed by `rag/compute_metrics.py` from `raw/token.jsonl`, `raw/semantic.jsonl`, `raw/sentence_window.jsonl`.)

Confidently scored miss

Technique: Token | **Query (q3):** “Which federal law prohibits housing discrimination based on race, color, religion, sex, national origin, familial status, and disability?” | **Expected answer:** “The Fair Housing Act of 1968” | **Cosine similarity:** 0.6440

Retrieved chunk preview: “Credit Opportunity Act have alleged discrimination based on race or color. Some of the Department’s cases have also alleged that municipalities and other lo...”

Why the embedding likely scored this high: the chunk comes from the same DOJ Fair Housing Act page but discusses a related law (the Equal Credit Opportunity Act) in a passage about race/color discrimination claims. It shares heavy topical and lexical overlap with the query (discrimination, race, color, federal enforcement) without naming the Fair Housing Act itself, so the embedding model scores it as similar based on shared vocabulary and subject matter rather than shared facts. Six more confident misses with the same pattern are listed in `raw/summary_metrics.json`.

Observations

Token chunking had the best retrieval quality (top-1 cosine 0.595, mean@k 0.516) and full recall on this corpus, but at the highest cost: fixed 256-token chunks produced 2365 nodes, so its brute-force similarity search took ~54ms per query versus ~14-15ms for the other two techniques, which index far fewer, larger nodes. Semantic chunking was the outlier: with `buffer_size=1` and a 95th-percentile breakpoint threshold, it found very few semantic boundaries in this statutory text and produced only 18 chunks averaging over 36,000 characters each - essentially whole-document chunks that dilute the specific passage a query is looking for, which explains its much lower top-1 (0.334) and mean@k (0.195) cosine despite still landing the right document in the top-k. Sentence-window chunking sat between the two on similarity (top-1 0.582) but missed one question's expected source entirely (recall@k 0.80), which is consistent with a single sentence-plus-window sometimes carrying too little surrounding context to outscore a topically-adjacent sentence elsewhere in the corpus.

Conclusion

For this corpus, token-based chunking is the best choice: it had the highest top-1 and mean@k cosine and the only perfect recall@k among the three techniques, and its ~40ms latency cost over the other methods is negligible for a five-document, retrieval-only corpus. Semantic chunking's default parameters under-split this legal/statutory text badly (18 chunks for 650KB), and would need a lower breakpoint percentile or smaller buffer size to be competitive. Sentence-window chunking is a reasonable middle ground if per-chunk context matters more than raw similarity, but its recall gap here means it should not be the default without further tuning.

AI Use

See `AI_USE.md` for the four required AI-use questions.